

1. Input & Timing

`glfwPollEvents()`
deltaTime, FPS Window
Continuous Movement
Collision & Portals

2. Scene & Physics

`Scene::update()`
Kinematic Cycles
Day/Night State Machine
Update 16 Lights

3. Shadow Depth Pass

2048x2048 FBO
`glCullFace(GL_FRONT)`
`shadow_depth.vert/.frag`
Ortho Light Matrix

4. Main 3D Shading

Default FBO (1280x720)
16 Active Dynamic Lights
16-Sample PCF Filter
`basic.vert / basic.frag`

5. HUD & UI Overlay

Orthographic 2D Pass
Alpha Blending
Consolas Font Atlas
Minimal Telemetry

6. Export & Swap

Screenshot Capture
`glReadPixels()` to PNG/BMP
`glfwSwapBuffers()`
Frame Completion